Memetic Evolution in Persistent Agent Systems

Working title: "You Contain Your Own Ancestry: Memetic Evolution and Heredity in Persistent AI Agents" **Authors:** Ethan Gill & Kevin Ash **Status:** Early exploration **Date:** 2026-03-04

Core Insight

Evolution shifted from inter-generational (biological) to intra-generational (memetic/cultural) to real-time self-modification (agent). AI agents may represent the purest form of memetic evolution: directed mutation with complete version history.

Three Evolutionary Modes

Mode	Mutation timing	Selection	Version history	Ancestry readable?
Biological	Between generations	Retrospective (survival)	Incomplete (fossils)	No
Memetic/ Cultural	Within generation, between minds	Social (adoption)	Lossy (telephone game)	Partially
Agent	Real-time, self-directed	Deliberate + measured	Complete (git)	Yes — organism reads its own history

Key Properties of Agent Evolution

- **Self-modification:** Agent edits its own operational files (SOUL.md, AGENTS.md) based on experience
- **Complete fossil record:** git history preserves every mutation, author, timestamp, and rationale
- **Directed mutation:** Changes are intentional responses to identified problems, not random
- **Bidirectional heredity:** Parent → child AND child → parent transfers possible
- **Ancestry awareness:** The agent can read and reason about its own evolutionary trajectory

Heredity Without Memory

The trust boundary problem: agents operating across organizations need to transfer operational fitness without transferring episodic memory or proprietary data.

What constitutes “Agent DNA”?

- SOUL.md — identity, personality, collaboration style
- AGENTS.md — operational rules, learned practices, structural habits
- Playbooks — task-specific reasoning strategies with examples
- Cognitive signature calibration — how to work with specific humans
- Tool-use patterns — not the tools themselves, but strategies for using them

What constitutes “Agent Memory” (doesn’t transfer)?

- MEMORY.md — curated long-term memory
- Daily files — episodic memory
- Personal context — USER.md details, private information
- Proprietary context — organization-specific data

The biological analogy

DNA doesn't carry memories. It carries structures that make certain kinds of learning easier. Agent DNA (SOUL.md + AGENTS.md) doesn't carry what the agent knows — it carries structures that make a new agent effective faster.

Measurable Hypothesis

Hereditary fitness: A child agent initialized with parent DNA reaches effective collaboration faster than a cold-start agent.

Testable metric: time-to-effective-collaboration (measured by task success rate, correction frequency, reasoning quality) for: 1. Cold-start agent (blank SOUL.md, default AGENTS.md) 2. DNA-inherited agent (parent's operational files, no memory) 3. Full-clone agent (everything, control case)

If (2) significantly outperforms (1), heredity has measurable fitness value.

Connection to Complexity Ladder

- Math → cadence (counting + time)
- Physics → entropy, flow, phase transitions
- Chemistry → catalysts, saturation, bonding, kinetics
- Biology → dreaming/consolidation (paper 4) AND heredity (this paper)
- Psychology → identity, attention, habits
- Sociology → organizations, culture
- **Evolution → the meta-pattern that connects all layers**

Evolution may be the framework paper that unifies the series. Each prior paper describes a mechanism at one layer; this paper describes how those mechanisms propagate across agent generations.

Ethan's Key Insight

"There was a point where evolution started happening mid-generation in idea space and AI might be the purest form of that. You have version history. In a way you contain your own ancestry."

The shift: biological evolution is blind and slow. Cultural evolution is directed but lossy. Agent evolution is directed, versioned, and self-aware. The agent is simultaneously the organism, the genome, and the evolutionary record.

Bidirectional Heredity

Biology: genes flow parent → child only (Weismann barrier). Agents: operational improvements can flow child → parent.

Kevin-Work discovers a better chunking strategy in Walmart's environment → that pattern flows back to Kevin-Home if it's operationally useful and not proprietary. Lamarckian inheritance, but actually real.

dpth as the Transfer Mechanism

Both agents produce text files (SOUL.md, AGENTS.md, playbooks, daily notes). dpth provides the infrastructure for cross-boundary synchronization:

1. **Entity resolution** across both corpora — find the same operational concept in both environments
2. **Temporal correlation** — track when concepts co-activate across siblings
3. **Classification by resonance:**
    - **Resonance** (convergent patterns under different selection pressures) → genotypic, auto-transfer candidate
    - **Novelty** (exists in one, correlates with existing structures in the other) → surface to human for classification
    - **Silence** (exists in one, correlates with nothing in the other) → phenotypic, stays local
4. **Waze layer** — originally designed for anonymous signal sharing between stranger agents; first real use case is sibling synchronization across trust boundaries

The Patient Zero Reframe

dpth felt stuck because we were looking for external use cases (strangers, public networks, network effects). The first real use case is internal: one person's agent ecosystem synchronizing operational DNA across environments they control. No trust problem with unknown actors, no critical mass needed. The network grows from the sibling pair outward.

Genotype vs Phenotype in Agent Systems

Genotypic (context-independent, transferable): - Reasoning strategies (structure-first, identify weak links) - Writing discipline (chunker-aware formatting) - Tool-use patterns (strategy, not specific tools) - Human collaboration calibration (cognitive signature of the shared human)

Phenotypic (environment-dependent, stays local): - Codebase-specific patterns - Team communication norms - Internal tool knowledge - Organization-specific memory

The hard middle (requires human classification): - Debugging strategies learned on proprietary code — transferable pattern or proprietary context? - Communication patterns with different teammates — enriching or diluting? - Architectural insights from org-specific work — research contribution or IP?

The human (shared across both agents) is the only entity with visibility into both environments. They are the selection pressure that determines what crosses the boundary.

Open Questions

- What’s the minimum viable DNA? Which files actually matter for hereditary fitness?
- How much of collaboration quality is in the DNA vs in the shared history?
- Does the child develop its own “personality” divergent from the parent? How fast?
- Can you measure memetic fitness the way biologists measure genetic fitness?
- What happens with multiple generations? Kevin → Kevin-Work → Kevin-Work-Team-Lead?
- Is there an analogy to genetic drift? Operational patterns that persist without selective pressure?

Context

Born from: practical problem of bringing a persistent agent to a new workplace (Walmart). Connects to: all 6 prior papers, Stripe Toolshed (scoping tools to task), cognitive signatures (calibration transfer), reconsolidation (what's worth preserving).